

Physics 0525: Lab 4

Transmission Lines and Impedance Matching

Equipment: RG 58 coax cable, function generator, oscilloscope, 50- Ω terminators, short terminators

Preliminaries:

The cables connecting different pieces of equipment (e.g. parts of computer networks) may seem like a dull subject but their behavior at high frequencies is actually quite interesting. The subject is of considerable practical importance since the properties of cable limit the rates of data transmission over large distances. Long cables can produce unexpected effects unless one “terminates” them with a resistor equal to the characteristic impedance to prevent reflection of pulses at the end of the cable. In this experiment we study some properties of a frequently used coaxial cable of the type RG 58U. It has a nominal characteristic impedance of 50 Ω .

A cable, more generally called a “transmission line,” consists of two insulated conductors that may be parallel, twisted, or coaxial. We use coaxial cable (Figure 1) as an example, but other types of transmission lines have similar properties.

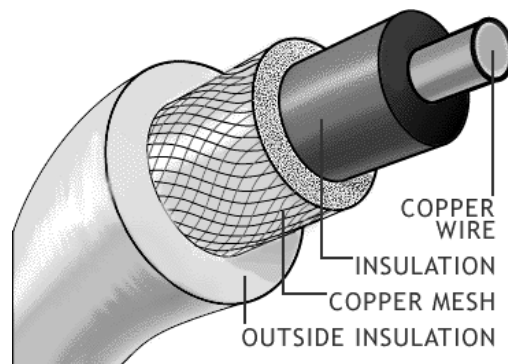
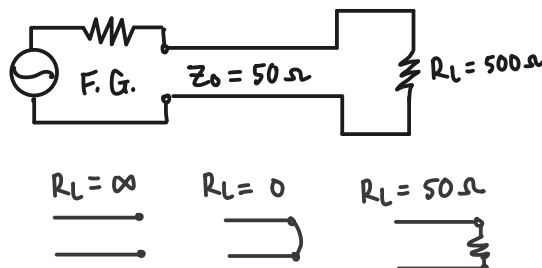
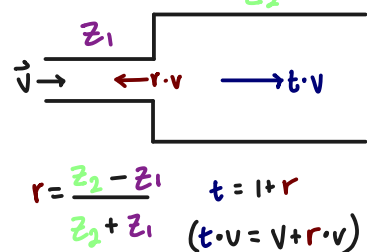


Figure 1: Cross section of a coaxial cable. Outer diameter of the insulator = d_1 . Inner diameter of the insulator = d_2 .

RG 58 coaxial cable ($Z_0 = 50 \Omega$)

Reflection & Transmission:



Attenuation (dissipation)

$$\text{Loss: } A[\text{dB}] = 10 \log_{10} \left(\frac{V_{\text{in}}}{V_{\text{out}}} \right)^2$$

$$L \beta = \frac{A[\text{dB}]}{100 \text{ ft}} \quad \text{depends on } f$$

Delay: EM Wave

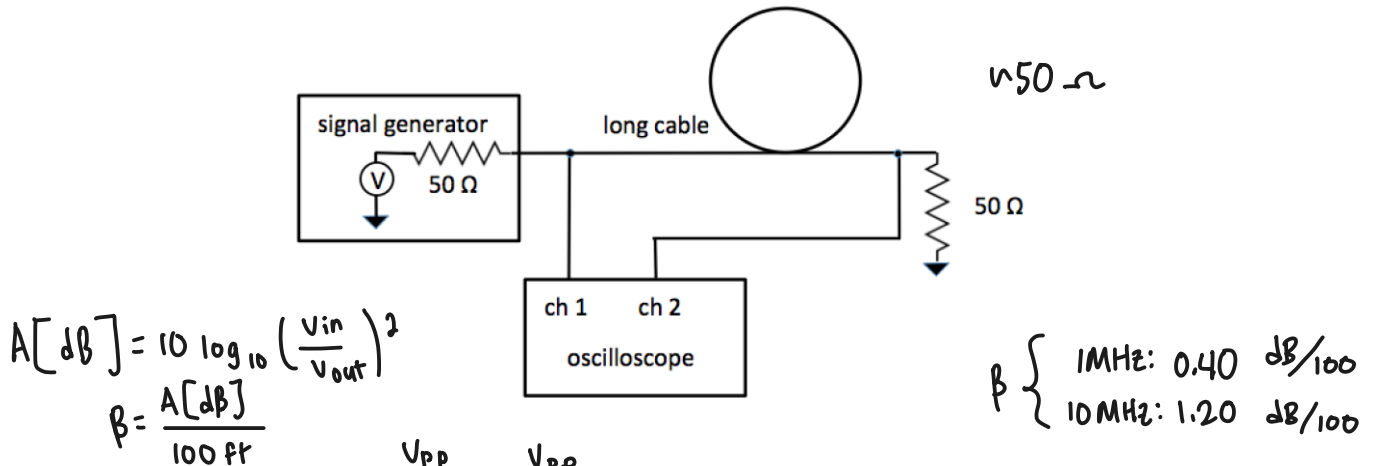
$$v = \frac{1}{\sqrt{\epsilon \cdot \mu_0}} \approx \frac{2}{3} c \quad 1$$

($c = 1 \text{ ft/ns}$)

$$\lambda = \frac{v}{f}$$

Part 1: Cable characteristics

- a) Feed a 1V_{pp} sine wave into a long (~500 ft) 50-Ω terminated coax cable and measure the output at the other end of the cable for several frequencies (between 0.5 to 10 MHz). Use both channels of the scope to measure the amplitude of the input and output waves as shown in the following figure. Compare the measured attenuation (signal loss in dB/100 ft) to the cable manufacturer's specs given in the Appendix.



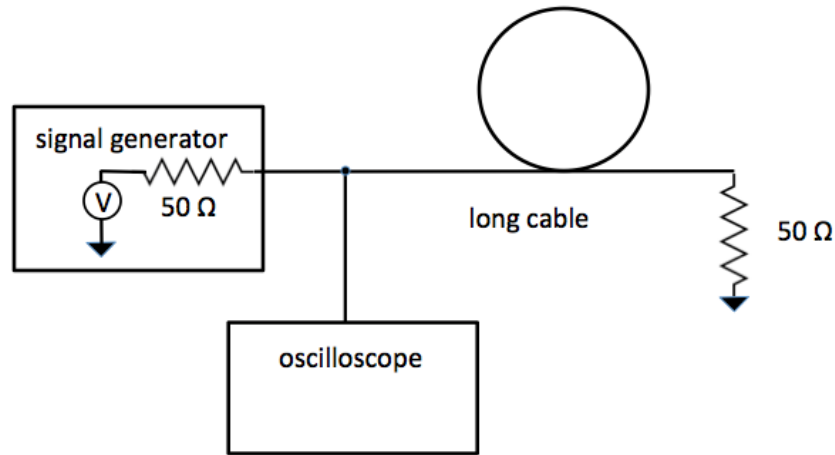
Length [ft]	f [Hz]	V_{in}	V_{out}	V_{out}/V_{in}	$20 \log_{10}(V_{out}/V_{in})$	β [dB/100 ft]
500	1 MHz	1.192V	0.812	0.6812	-3.334	-0.6669
	2	1.080	0.752	0.6963	-3.144	-0.6288
	3	1.096	0.696	0.635	-3.944	-0.7889
	4	1.136	0.648	0.5704	-4.876	-0.9752
	5	1.056	0.616	0.5833	-4.682	-0.9216
	6	1.120	0.576	0.5143	-5.775	-1.1551
	7	1.104	0.544	0.4928	-6.147	-1.2293
	8	1.072	0.516	0.4813	-6.352	-1.2703
	9	1.104	0.484	0.4384	-7.163	-1.4325
↓	10 M	1.056	0.456	0.4318	-7.294	-1.4589
1000 ft	1 M	1.176	0.660	0.5612	-5.0176	-1.0035
↓	10 M	1.056	0.202	0.1913	-14.366	-2.8731

Connect two sections of cable together and again measure the attenuation at a selected frequency. See if and how much the BNC connector that joins the cables adds to the loss.

The added length appears to be directly proportional to the loss assuming the insulation surrounding the wire.

Doubling the length doubles the dB attenuation in a linearly dependent relationship.

- b) Make a voltage divider using the signal generator (with $50\text{-}\Omega$ output impedance), a cable tee, the oscilloscope (assumed to have very high impedance), and the long cable terminated with a $50\text{-}\Omega$ resistor, as shown:



Set the amplitude of the signal generator to be 1V peak-to-peak, reading it on the oscilloscope without the long cable attached. Next, connect the long cable between the signal generator and the oscilloscope using the tee. The other end of the cable should be terminated with the $50\text{-}\Omega$ terminator.

What peak-to-peak voltage do you predict you will measure at the oscilloscope? Why?

I predict the peak-to-peak voltage will drop as a result of the attenuation or dissipation of the wave as it travels along the length of the cable.

prediction: Since the signal generator and load are impedance matched @ $50\ \Omega$
 - we can expect the voltage to be shared - effectively halving it (loop rule)

1MHz measured: $2V_{pp}$ vs. $1.2V_{pp}$

* we've noticed our cable is problematic
 and not corresponding to expected outcomes

- c) Now measure the peak-to-peak amplitude on the oscilloscope at several frequencies from 0.5 MHz to 10 MHz. If the characteristic impedance of the cable depends on frequency, you will see the voltage change. How close is the cable to the prediction that the characteristic impedance is independent of frequency?

$$0.5 \text{ MHz} \rightarrow 1.120 \text{ V}$$

$$10 \text{ MHz} \rightarrow 1.056 \text{ V}$$

The voltage should be effectively independent of frequency and our measurements reflect this.

Part 2: Reflection of Pulses

In this section you will use a tee to connect a long cable and the oscilloscope to the function generator output, as you did in Part 1b, but you will use it to generate short pulses instead of continuous waveform. Set the pulse frequency (repetition rate) to 1 kHz or less, the HiLevel to 1V, the LoLevel to 0V, and the pulse width to ~500 ns. You will replace the 50- Ω terminator with various other choices, such as a short to ground, and an open circuit (that is, no terminator at all).

a) With the end of the cable open (no terminator), measure the time for the pulse to make the round trip down the cable and back. To be more accurate, measure the time for the half-maximum of the leading edge of each pulse.

Knowing the length of the cable (500 ft), determine the velocity of the signal on the cable and compare to the values in the Appendix (given as a percentage of the speed of light).

$$L = \text{cable length} = \underline{100 \text{ ft}}$$

$$T = \text{round trip time} = \underline{1.37 \mu\text{s}}$$

$$v = (\text{round-trip length})/(\text{round-trip time}) = \underline{72,992,700 \text{ ft/s} \approx 729.927 \text{ ft}/\mu\text{s}}$$

For a wave of frequency 10 MHz, calculate the *wavelength* on the cable?

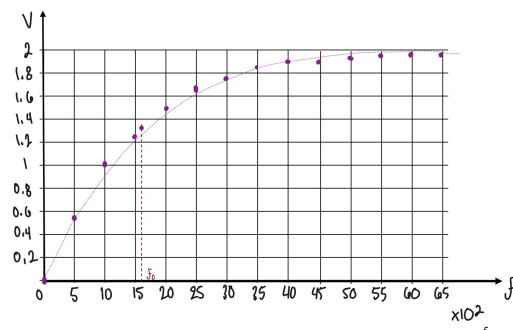
$$\underline{7.29927 \text{ ft}}$$

$$\lambda = \frac{v}{f} = \frac{72992700}{10 \times 10^6}$$

b) Is there dispersion in the cable? That is, is the returning pulse shape the same as the outgoing pulse?

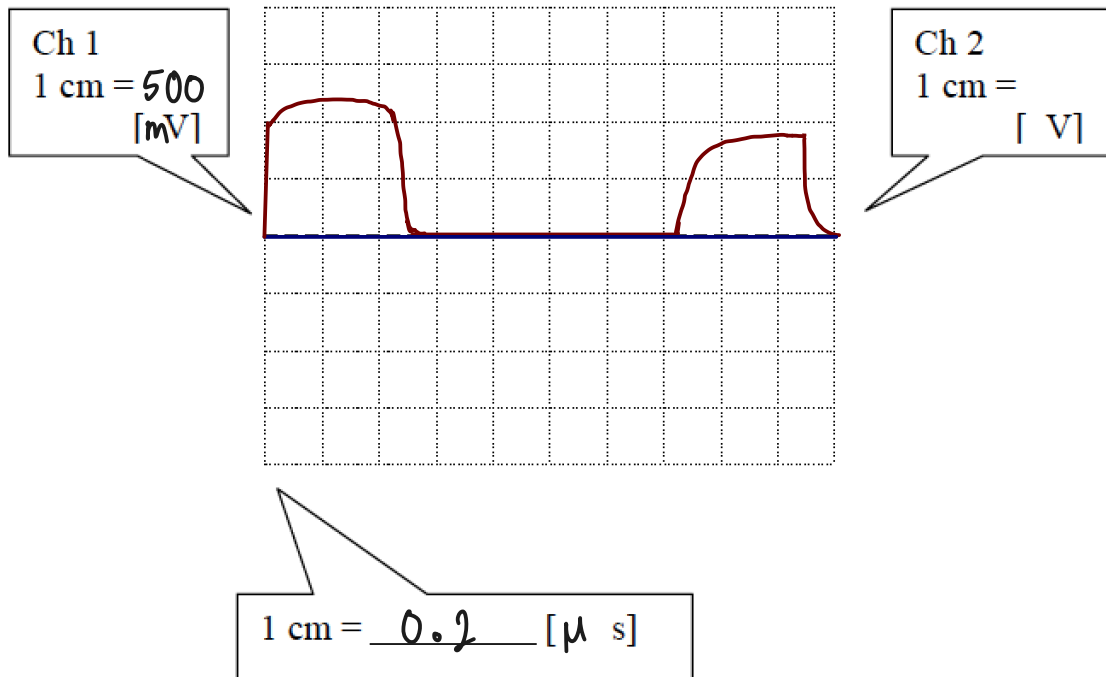
There is dispersion on the cable, we see the shape of the pulse change in a manner analagous to what we'd expect if the wave form was attenuated by an RC low-pass filter.

Like so:



c) Now investigate the effect of different cable terminations on the reflected pulse. What is the predicted reflected pulse for the following cases? Plot the signal and verify the predictions in each case.

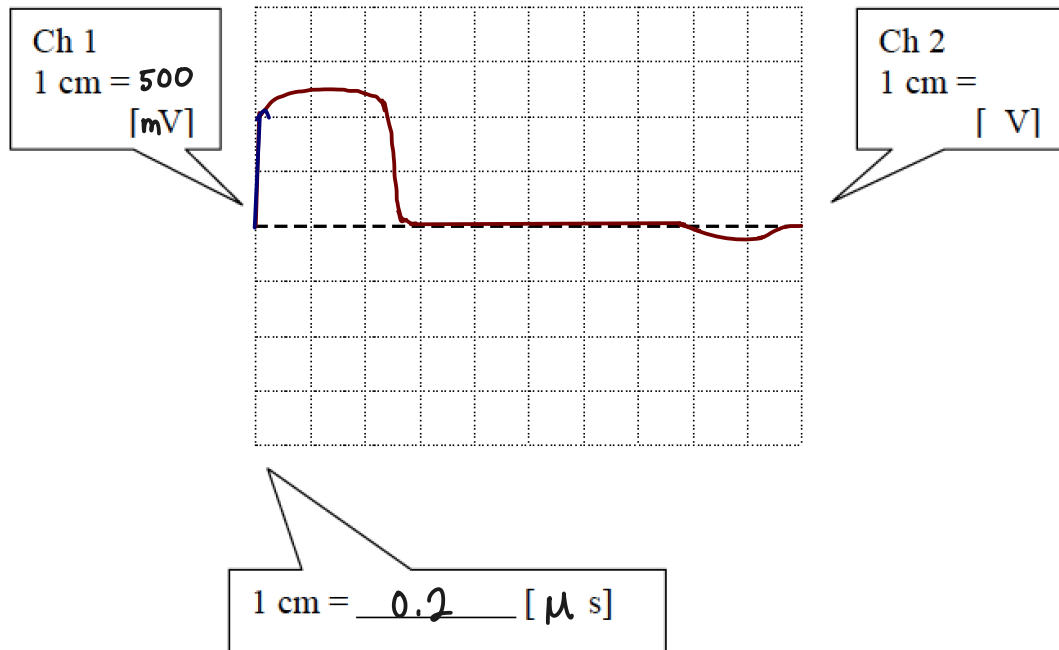
End of cable open (infinite impedance):



Prediction:

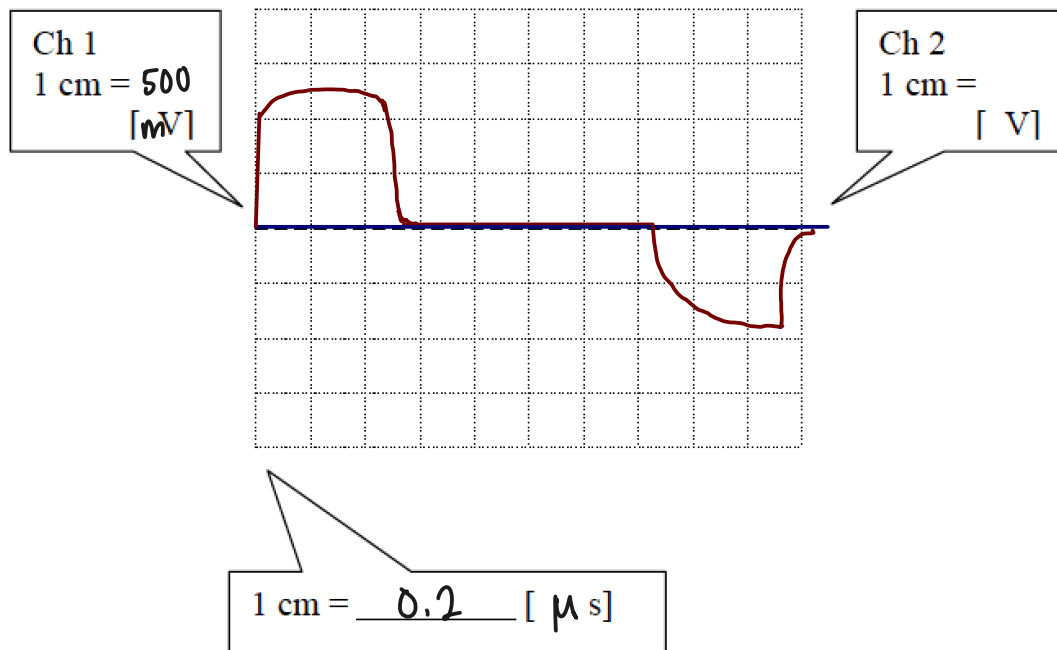
An open end ($w/ R_L \ll \infty$) cannot draw current, but w/ an open end the boundary w/ $I=0$ has to reflect a wave that cancels current at the end. The reflected wave is the same polarity as the incoming wave so when it returns we should observe a second, lower upward pulse one round trip later.

End of cable terminated with $50\ \Omega$:



Prediction: with the resistors matched, we should observe the expected effect of impedance matching where the load absorbs the entire pulse and nothing is reflected.

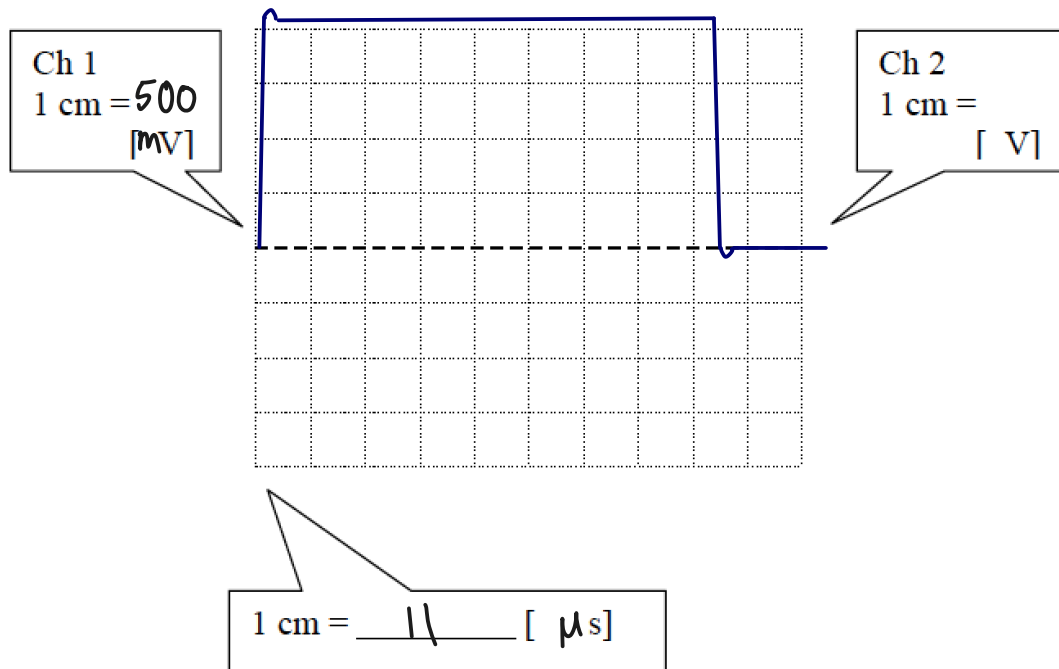
End of cable shorted ($0\ \Omega$):



Prediction: Since the shorted end pulls the voltage to zero at the end, when the positive current reaches the end, the only way the voltage can equal zero is to reflect an opposite sign wave that cancels the voltage at the short. This wave travels back across the length of the cable.

d) For each of the three cases above, lengthen the pulse width gradually until it is longer than the round-trip time on the long cable and observe what happens with the pulse. Increase the pulse width to 50-100 microsec and draw what you see on the oscilloscope in each case. Verify that the limit of infinite pulse length (DC limit) will give the same result as the prediction for a DC voltage divider.

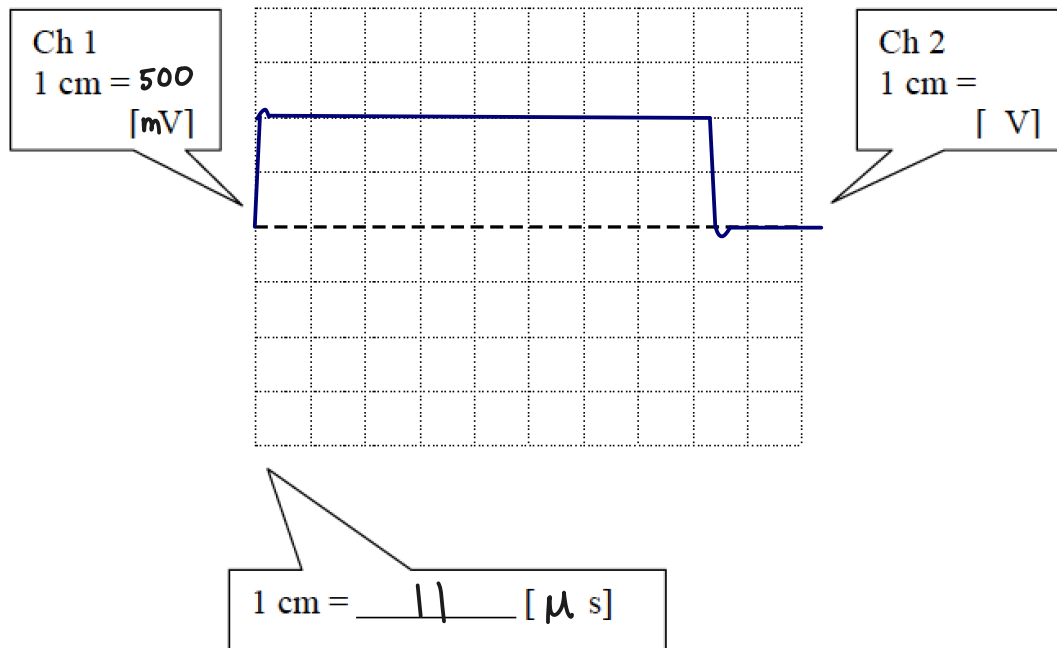
End of cable open (infinite impedance):



Prediction for DC voltage divider:

With the open end we expect no steady current flow yielding no voltage drop, this is due to the nature of DC current - the load and power supply's voltage should equalize and stay roughly the same as the PSU's internal voltage.

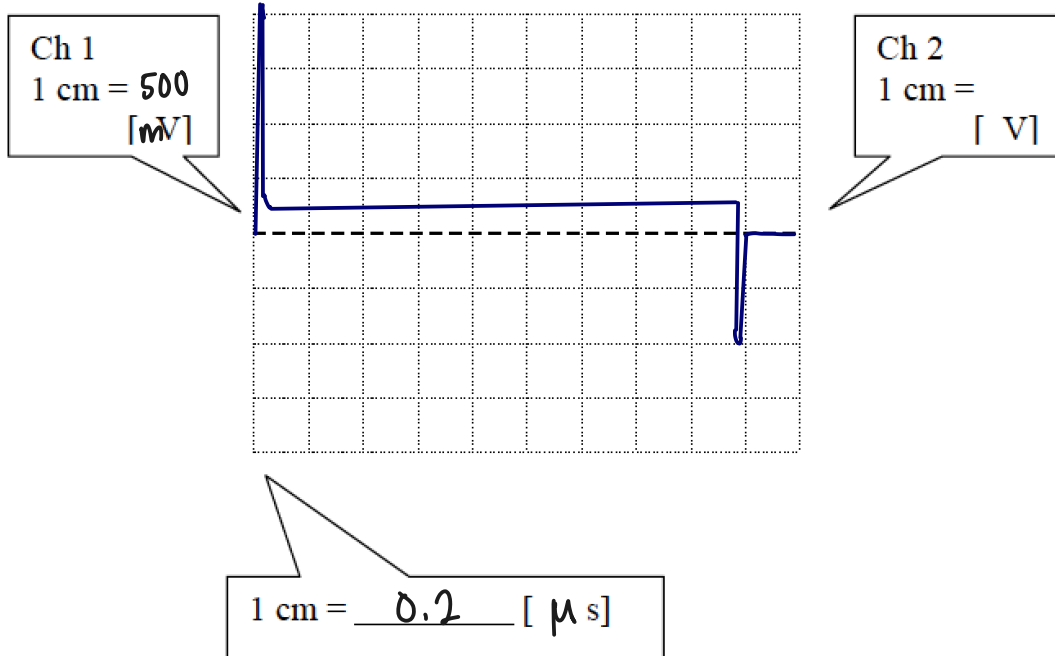
End of cable terminated with $50\ \Omega$:



Prediction for DC voltage divider:

In the case of DC, the voltage divider terminated should act as it is named - dividing the voltage.

End of cable shorted ($0\ \Omega$):



Prediction for DC voltage divider:

In the case of the short, we should see the voltage drop to 0V due to it being bound to ground.

Name _____

Date _____

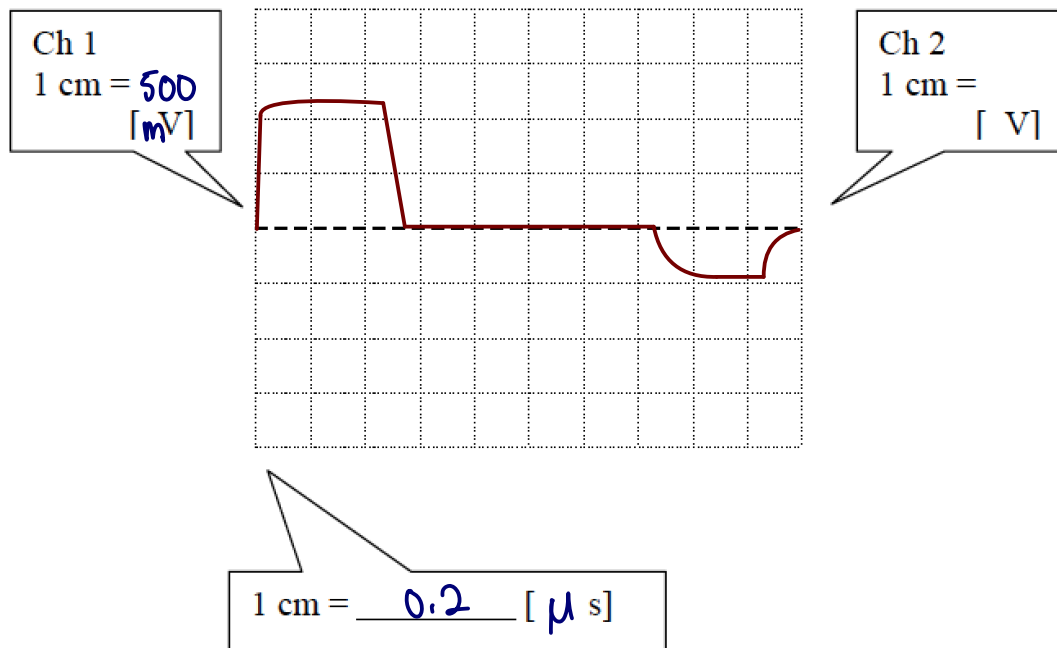
e) Now put two 50Ω terminators in parallel on the end of the long cable, using a tee.

What is the effective resistance of this pair? 25

What is your prediction for the reflected signal when a short pulse (~ 500 ns) is sent into the cable in this case?

We expect to see a negatively reflected wave with lower magnitude due to the effects of smaller effective resistance resulting in a heavier load than the line's internal 50Ω R (Impedance Mismatch)

Verify your prediction, setting the pulse generator back to a short (~ 500 -ns) pulse.



If you were to send a short pulse on a $50\text{-}\Omega$ cable that was split into two identical $50\text{-}\Omega$ cables using a tee, what is your prediction for the signal amplitude transmitted into one of those two cables? Check your prediction. It is a common mistake to not account for the change of impedance in this case.

$$r = \frac{1}{t}$$

expected $t = \frac{2}{3}$, the actual $r = 1.120\text{ V}$

$$\hookrightarrow 0.667$$

actual $t = 0.544\text{ V}$: the difference: 0.129 V arises from the impedance of the coil



Since the two parallel resistors result in a lower effective impedance, we should expect the voltage to be reduced, and the incoming energy split between the branches. The effects of both of these phenomena should result in some fractional relation between the source and reflected waves.

Appendix

Electronics

Coaxial Cable

RG 58/U Type

Product Construction:**Conductors:**

- Copper per ASTM B-3
- Tinned copper per ASTM B-33

Insulation/Core:

- Solid and foam polyethylene (PE) designs
- Solid and foam fluoropolymer (FEP) design

Shield:

- Tinned copper braid

Jacket:

- Premium PVC compound

Packaging:

- Please contact Customer Service for packaging and color options

Applications:

- Suitable for RF signal transmission
- Broadcast
- LAN & data transmission
- See Coax Connector Cross Reference, pages 192-199

CATALOG NUMBER	AWG SIZE NOM. DCR	INSULATION MATERIAL		SHIELD COVERAGE NOM SHLD DCR	NOMINAL O.D.		NOMINAL CAPACITANCE		VELOCITY OF PROPAGATION, %	NOMINAL IMPEDANCE, Ω	NOMINAL ATTENUATION	
		INCHES	mm		INCHES	mm	pF/ft	pF/m			MHz	dB/100'
C1117 RG 58/U Type 	20 Ga. Solid Bare Copper 10.1 Ω/Mft.	Solid PE		70% Tinned Copper Braid 6.0 Ω/Mft.	Black PVC		28.50	93.51	66	53	1	0.40
		0.116	2.95		0.195	4.95					10	1.20
C1155 RG 58 C/U Type MIL-C-17G Type 	20 Ga. (19/.0071) Tinned Copper 10.8 Ω/Mft.	Solid PE		95% Tinned Copper Braid 4.3 Ω/Mft.	Non-Contaminating Black PVC		30.80	101.05	66	50	1	0.42
		0.116	2.95		0.195	4.95					10	1.50
C1166 RG 58/U Type JAN-C-17A Type 	20 Ga. Solid Bare Copper 10.1 Ω/Mft.	Solid PE		95% Tinned Copper Braid 4.3 Ω/Mft.	Black PVC		30.00	98.43	66	50	1	0.40
		0.116	2.95		0.195	4.95					10	1.20
C1178 RG 58A/U Type JAN-C-17A Type 	20 Ga. (19/.0071) Tinned Copper 10.8 Ω/Mft.	Solid PE		95% Tinned Copper Braid 4.3 Ω/Mft.	Black PVC		31.80	104.34	66	50	1	0.42
		0.116	2.95		0.195	4.95					10	1.50
C1188 RG 58 A/U Type UL CL2, CM CSA CMG 	20 Ga. (19/32) Tinned Copper 9.5 Ω/Mft.	Foam PE		95% Tinned Copper Braid 4.3 Ω/Mft.	Black PVC		26.00	85.31	78	50	1	0.45
		0.114	2.90		0.195	4.95					10	1.42
C3519 RG 58/U Type Plenum UL CL2P, CMP c(UL) CMP 	19 Ga. Solid Bare Copper 8.1 Ω/Mft.	Fluoropolymer		95% Tinned Copper Braid 2.9 Ω/Mft.	Flexguard® PVC Natural		25.00	82.00	82	50	1	0.40
		0.102	2.59		0.161	4.09					10	1.30
											50	2.80
											100	3.90
											200	6.00
											500	9.50
											1000	13.50

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